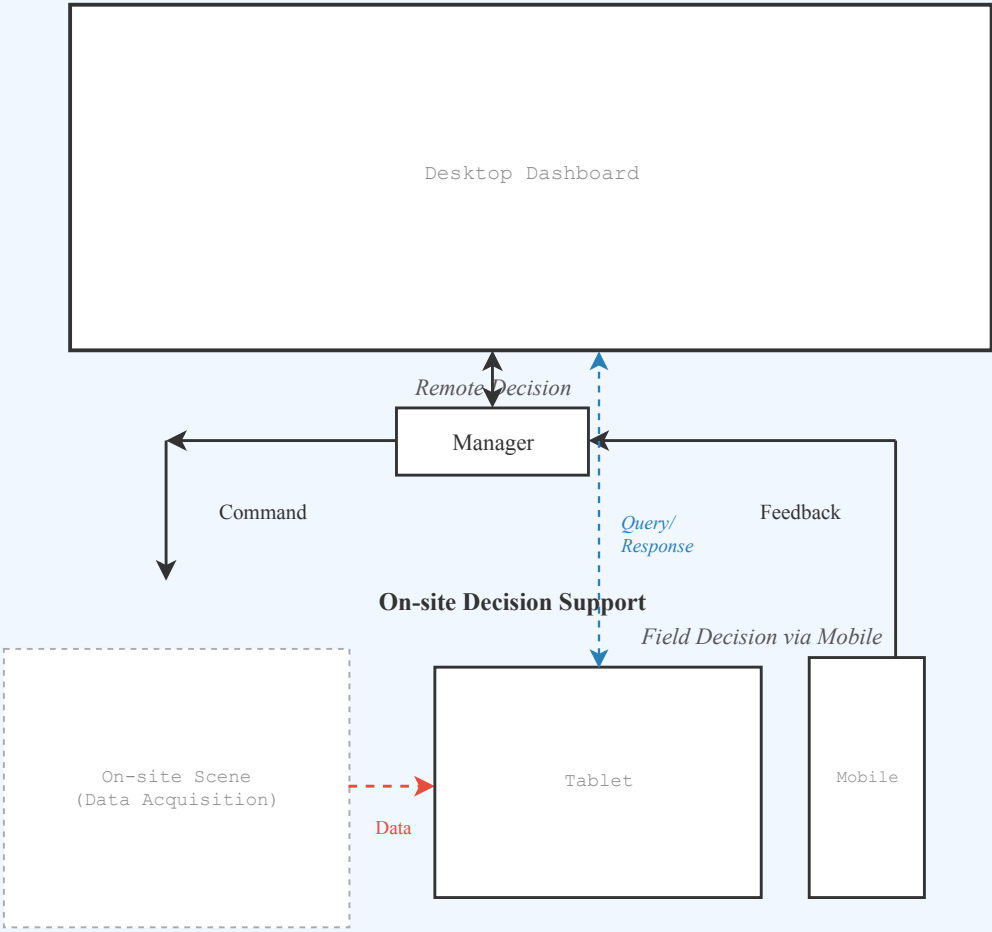


(a) BUILDING DEFECT MANAGEMENT

VALIDATED

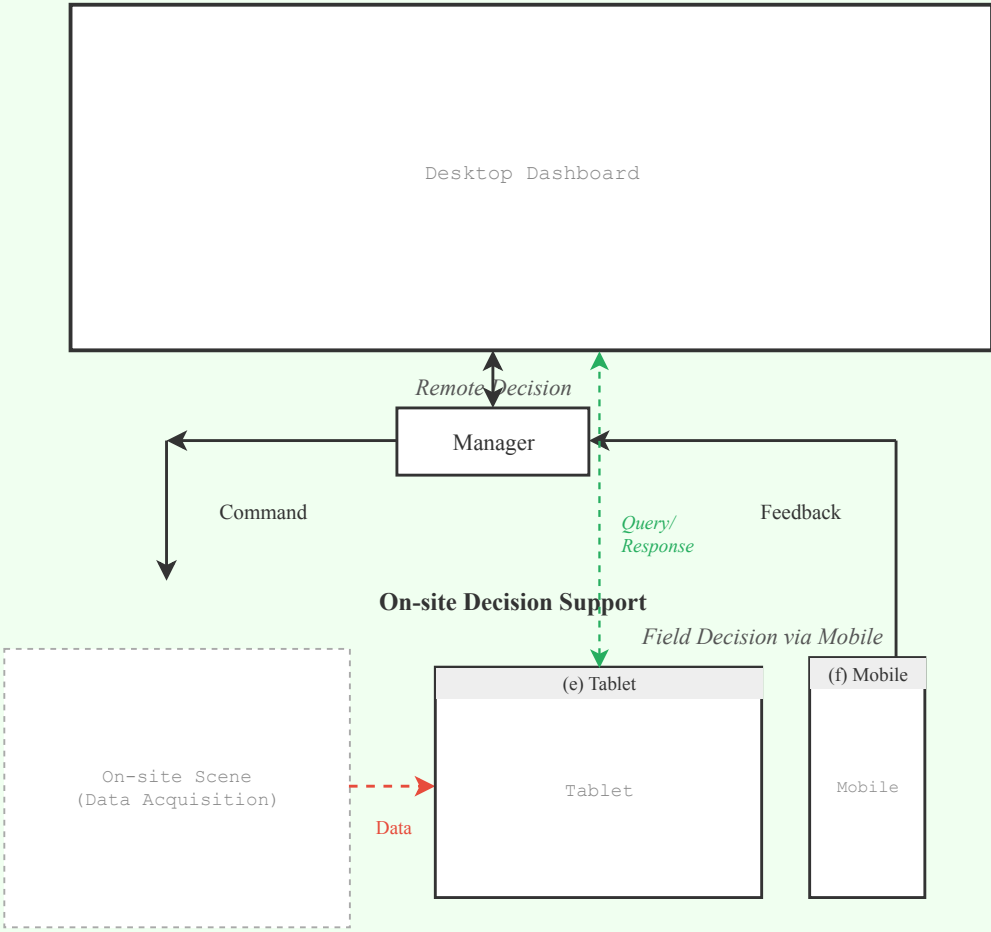
RAG + LLM Decision Support Platform



(b) CONSTRUCTION PROGRESS MANAGEMENT

CONCEPTUAL

RAG + LLM Decision Support Platform



Framework Transfer: Invariant Core Components

EP $\langle \mathcal{O}, \mathcal{G}, \mathcal{D} \rangle$
Observations, Geometry, Dependencies

DP $\langle \mathcal{I}, \mathcal{E}, \mathcal{S}, \mathcal{A} \rangle$
Identity, Evidence, Semantics, Analysis

k-hop $\mathcal{N}_k(\mathcal{A}_i)$
Context-aware retrieval

LLM + RAG Pipeline
Topology-aware reasoning

Adaptation: Defect: IDP→IAP, BIM Construction: WDP→WZP, Schedule Schema instantiation, not architectural modification